

David Ibarra Luna

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SUMMARY

Industrial Chemist with a Master's in Materials Science, specialized in nanomaterials for optoelectronic water sensors applied to industrial water quality assurance. Experienced in interdisciplinary research combining materials development, contaminant analysis, and data processing with Python, aimed at solving real-world environmental challenges.

PROFESSIONAL EXPERIENCE

Research stay at institutes of CSIC (Barcelona, Spain) Sep 2024 - Dec 2024

- Developed carbon nanomaterials for the detection of nitrates in water, contributing to a project with direct applications in optoelectronic sensors.
- Applied advanced material characterization (electron microscopy, spectrophotometry) and processed spectral data with Python, expediting result interpretation.

Research Assistant - UANL, Faculty of Chemical Sciences Aug 2021 - Jun 2025

- Synthesized diverse nanomaterials (N-GQDs, AuNPs, carbon nanotubes, etc.), specializing in N-GQDs for thesis research.
- Performed polymer synthesis (e.g., Nylon) and synthesized ionic liquids to support parallel research lines within the department, and employed various analytical and purification techniques, including Soxhlet distillation for extraction and gravimetric methods for molecular weight determination.

Laboratory Analyst - UANL, Dept. of Food, Toxicology, and Drugs Jun 2020 - Jan 2021

- Performed quantitative analysis of food samples to determine nutritional content, ensuring data accuracy through strict adherence to laboratory protocols.

Robotics teacher (High School) Aug 2023 - Jun 2025

- Taught robotics and programming to high school students using the STEM-based FIRST Lego League methodology, fostering technical and problem-solving skills.

PROJECTS

Spectrum Data Loader - Python Package (PyPI) [Link to Project](#)

Developed and published a Python library to simplify the loading and initial processing of spectroscopy data files.

EDUCATION

2023–2025 M.Sc. in Materials Chemistry, Autonomous University of Nuevo Leon, (Grade: 94.4/100)

2019–2022 B.Sc. in Industrial Chemistry, Autonomous University of Nuevo Leon, (Grade: 82.7/100)

PUBLICATIONS

Ibarra, David et al. (Apr. 2023). “Synthesis of Graphene Quantum Dots Coupled to Au Nanoparticles: A Facile and Versatile Route Using Different Carbon Sources”. In: *C* 9.2, p. 45. DOI: [10.3390/c9020045](#).

SKILLS

Lab & Software Nanomaterial & Organic Synthesis, Material Characterization, Quantitative Analysis, Python (Pandas, NumPy, Matplotlib, ManimCE, Flask), Git, $\LaTeX$, ImageJ.

Languages Spanish (Native), English (Advanced), French (Beginner).